WONDMAGEGN TAFESSE TIRKASO

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RESEARCH INTERESTS

Environmental and climate economics • Agricultural productivity and technical efficiency • Energy economics and sustainable development • Spatial econometrics and spillover effects • Climate-agriculture interactions and adaptation • Antimicrobial resistance economics • One Health economics • Applied impact evaluation • Computable general equilibrium modeling

EDUCATION

PhD, Economics, Swedish University of Agricultural Sciences, Uppsala, Sweden (2014–2019)

Visiting PhD Student, University of Western Australia, Perth, Australia (June–August 2018)

MSc, Economics, Swedish University of Agricultural Sciences, Uppsala, Sweden (2010–2014)

MSc, Rural Development, Swedish University of Agricultural Sciences, Uppsala, Sweden (2009–2011)

BA, Economics, Jimma University, Jimma, Ethiopia (2004–2007)

ACADEMIC AND PROFESIONAL POSITIONS

Economist, United Nations Food and Agriculture Organization, Rome, Italy (September 2021–Present)

Researcher, Swedish University of Agricultural Sciences, Department of Economics, Uppsala, Sweden (January 2019–December 2020)

Doctoral Researcher, Swedish University of Agricultural Sciences, Department of Economics, Uppsala, Sweden (August 2014–December 2018)

Research Assistant, Swedish University of Agricultural Sciences, Department of Economics, Uppsala, Sweden (December 2013–May 2014)

PUBLICATIONS

Selected Peer-Reviewed Journal Articles

1) Tirkaso, W. (2026). On the impact of rural electrification: The role of complementary factors. *Journal of African Economies* (forthcoming).
2) Acosta, A., Tirkaso, W., et al. (2026). Antimicrobial use in livestock: The economic cost of action or inaction. *Journal of Agricultural Economics* (forthcoming).
3) Acosta, A., Tirkaso, W., Nicolli, F., Van Boeckel, T., & Song, J. (2025). Antimicrobial use in livestock: Future global perspectives. *Nature Communications*, 16, 2469.
4) Acosta, A., Nicolli, F., & Tirkaso, W. (2024). Cattle ownership and households' welfare: Evidence from Southern Africa. *Global Food Security*, 42, Article 100772.
5) Tirkaso, W.T. and Hansson, H. (2023). Assessing farm efficiency through quantities or revenues and costs – does it matter? *Journal of Agricultural and Applied Economics*, 55, 551-565.
6) Tirkaso, W. T., & Hailu, A. (2022). Does neighborhood matter? The effect of spatial proximity on farmers' technical efficiency. *Agricultural Economics*, 1–13.

7) Gren, I-M., and Tirkaso, W.T. (2021). Cost effective reduction in emissions of greenhouse gases – Fuel and food consumption and negative emissions in Sweden. *Energy Economics*, 105638
8) Marbuah, G., Gren, I-M., and Tirkaso, W.T. (2021). Social capital, economic development, and carbon emissions: Empirical evidence from counties in Sweden. *Renewable and Sustainable Energy Reviews*, 152,111691.
9) Tirkaso, W.T., and Gren, I-M. (2021). Evaluation of cost efficiency in hydropower related biodiversity restoration projects in Sweden: A stochastic frontier approach. *Journal of Environmental Planning and Management*, 1-20.
10) Tirkaso, W. T., & Gren, I.-M. (2020). Road fuel demand and regional effects of carbon taxes in Sweden. *Energy Policy,* 144, 111648.
11) Nsabimana, A., and Tirkaso, W.T. (2020). Examining coffee export performance in East African countries: Do Bilateral Trade Relations Matter? *Agrekon*, 1-19.

Manuscripts Under Review

- **Tirkaso, W. T.** & Ouedraogo, N. (2025). Spatial spillovers and household clean energy adoption in Africa: Evidence from Ethiopia, Nigeria, and Uganda. *[Under review in Energy Economics]*.
- Acosta, A., **Tirkaso, W.**, & Padungtod, P. (2025). Farm size and technical efficiency in Vietnamese pig production: Evidence from a time-varying stochastic frontier model. *[Under review in Agricultural Economics]*.
- Kangogo, D. & **Tirkaso, W. T.** (2025). Climate variability, conflict, and household welfare in Sub-Saharan Africa: A spatiotemporal analysis. *[Working Paper]*.

Policy Reports and Technical Briefs

- FAO. (2025). Antimicrobial use in livestock: The economic cost of action or inaction (In press). Food and Agriculture Organization of the United Nations. Contributors: Acosta, A., Song, J., **Tirkaso, W.**, Valcarce, A., Kabali, E., Dalton, R., Uwizeye, A., Cinardi, G., Araujo, S., Artavia, M., Chatzopoulos, T., Bullon, C., & Reantaso, M.
- **Tirkaso, W. T.**, Laderach, P., & Pacillo, G. (2021). How does climate exacerbate root causes of conflict in Uganda? An econometric analysis. International Center for Tropical Agriculture (CIAT), CGIAR FOCUS Climate Security, Factsheet 2021/14.
- Marbuah, G., **Tirkaso, W. T.**, & Gren, I.-M. (2016). Cost-effective policy and incomplete information on policy implementation. Towards a fossil-free society – in the Stockholm-Mälar region. Technical Report.

RESEARCH GRANTS AND AWARDS

- **Swedish University of Agricultural Sciences (SLU) – SLU Urban Futures** (July 2019)

  Competitive research grant supporting interdisciplinary analysis at the intersection of urban development, sustainability, and policy (SEK 48,000).
- **Swedish University of Agricultural Sciences (SLU) – NJ Faculty** (March 2019)

  Research grant supporting applied economic research and project development (SEK 76,700).
- **Swedish Foundation for International Cooperation in Research and Higher Education (STINT)** (January 2019)

  International mobility and collaboration grant supporting research cooperation and academic exchange (SEK 100,000).

- **SLU-FUR – Travel and Accommodation Grants** (Spring 2016, 2018)
  Competitive grants supporting international research dissemination and collaboration (SEK 64,000).
- **SLU Global – Travel Grant** (Autumn 2013)
  Grant supporting participation in international academic activities (SEK 2,000).

CONFERENCE PRESENTATIONS AND INVITED TALKS

- **High-Level Technical Workshop on Antimicrobial Use (AMU) Mitigation and RENOFARM Implementation.** Led technical presentations on FAO's microeconomic optimization model for AMU reduction. Lusaka, Zambia, August 19–22, 2025.
- **Capacity-Building Training on Applications of the Stochastic Frontier Model (SFM) Using Stata.** Organized by the General Statistics Office of Vietnam (GSO) and the Institute of Policy and Strategy for Agriculture and Rural Development (IPSARD). Hanoi, Vietnam, July 3–7, 2023.
- **First Meeting of the Quadripartite Technical Group on the Economics of Antimicrobial Resistance.** Organized by the World Health Organization (WHO). Geneva, Switzerland, April 4–5, 2023.
- **2019 AAEA Annual Meeting.** Agricultural and Applied Economics Association. Atlanta, Georgia, USA, July 21–23, 2019.
- **26th Ulvön Conference on Environmental Economics.** Organized by Umeå University and the Swedish University of Agricultural Sciences (SLU). Ulvön, Sweden, June 18–20, 2019.
- **EAAE Seminar 168: Behavioural Perspectives in Agricultural Economics and Management.** Organized by the Swedish University of Agricultural Sciences (SLU). Uppsala, Sweden, February 6–7, 2019.
- **2nd Nordic Annual Environmental and Resource Economics (NAERE) Workshop.** Organized by the Norwegian School of Economics (NHH). Bergen, Norway, April 12–13, 2018.
- **15th European Workshop on Efficiency and Productivity Analysis (EWEPA).** Organized by Loughborough University. Loughborough, United Kingdom, June 12–15, 2017.
- **Centre for Environmental and Resource Economics (CERE) Seminar.** Organized by Umeå University. Umeå, Sweden, October 25, 2017.
- **International Society of Ecological Economics (ISEE) 2016 Conference.** Organized by the University of the District of Columbia. Washington, D.C., USA, June 26–29, 2016.
- **Euroleague for Life Sciences (ELLS) Scientific Student Conference.** Organized by BOKU University of Natural Resources and Life Sciences. Vienna, Austria, November 14–16, 2013.

PROFESSIONAL TRAINING AND CERTIFICATION

Certificate in Data Science, University of California, San Diego (2020–2021)

Econometrics of Impact Evaluation, University of Copenhagen (June–August 2018)

Spatial Econometrics and Data Management, University of Copenhagen (August 2010–February 2014)

TECHNICAL SKILLS

Statistical and Econometric Software: Stata, R, Python (econometric modeling, data analysis, simulation methods, machine learning)

Spatial Analysis: ArcGIS, QGIS (spatial econometrics, geospatial data analysis, mapping)

Programming and Data Engineering: Python (web scraping, API integration, reproducible data pipelines, multi-country dataset integration)

Technical Writing: LaTeX, Microsoft Office (technical reports, policy briefs, scientific manuscripts)

Languages: English (fluent), French (basic), Swedish (basic), Amharic (native), Hadiyya (fluent)

REFERENCES

Available on request.